

Question 4 – Stage 1

A 5 by 15 by 27 cuboid is made up from 2025 unmarked wooden unit cubes.

Each face of the cuboid is assigned a different integer from 0 to 5 and that number is written on every unit square making up that face.

The 2025 cubes are separated.

The value of a unit cube is the sum of the numbers (if any) on its faces.

Given that the sum of all the numbers on all the faces is 2025, how many cubes have an odd value?

Working

Each face has an area equal to its number of unit squares.

Finding Possible Face Labels

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In [1]: from itertools import permutations

faces = [75,75,135,135,405,405]
labels = [0,1,2,3,4,5]

correctArrangements = []

for possibility in permutations(labels,len(labels)):
    pairs = list(zip(faces,list(possibility)))
    total = sum([i[0]*i[1] for i in pairs])

    if total == 2025:
        correctArrangements.append(pairs)

print("Possible Combinations:")
for comb in correctArrangements:
    print(comb)
```

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Possible Combinations:
[(75, 4), (75, 5), (135, 1), (135, 3), (405, 0), (405, 2)]
[(75, 4), (75, 5), (135, 1), (135, 3), (405, 2), (405, 0)]
[(75, 4), (75, 5), (135, 3), (135, 1), (405, 0), (405, 2)]
[(75, 4), (75, 5), (135, 3), (135, 1), (405, 2), (405, 0)]
[(75, 5), (75, 4), (135, 1), (135, 3), (405, 0), (405, 2)]
[(75, 5), (75, 4), (135, 1), (135, 3), (405, 2), (405, 0)]
[(75, 5), (75, 4), (135, 3), (135, 1), (405, 0), (405, 2)]
[(75, 5), (75, 4), (135, 3), (135, 1), (405, 2), (405, 0)]
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Meaning the faces must be:

$$5 \times 15 \in 4, 5$$

$$5 \times 27 \in 1, 3$$

$$15 \times 27 \in 0, 2$$

$$\begin{aligned} \text{number of Odd cubes} &= 5 \text{ cubes} + 1 \text{ cubes} + 3 \text{ cubes} + \text{odd overlaps} \\ &= (3 \cdot 13) + (3 \cdot 25) + (3 \cdot 25) + \text{odd overlaps} \\ &= 189 + \text{odd overlaps} \end{aligned}$$

$$\begin{aligned} \text{odd overlaps} &= (4,1 \text{ cubes}) + (4,3 \text{ cubes}) + (5,0 \text{ cubes}) + (5,2 \text{ cubes}) + (1,0 \text{ cubes}) + (1,2 \text{ cubes}) + (3,0 \text{ cubes}) + (3,2 \text{ cubes}) + \text{odd corners} \\ &= 3 + 3 + 13 + 13 + 25 + 25 + 25 + 25 + \text{odd corners} \\ &= 132 + \text{odd corners} \end{aligned}$$

$$\begin{aligned} \text{odd corners} &= (4,1,0 \text{ cube}) + (4,1,2 \text{ cube}) + (4,3,0 \text{ cube}) + (4,3,2 \text{ cube}) \\ &= 4 \end{aligned}$$

$$\begin{aligned} \text{number of Odd cubes} &= 189 + 132 + 4 \\ &= 325 \end{aligned}$$